



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

ARTIFICIAL INTELLIGENCE [IT-4401]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) While designing an AI tutor. **Analyze** how inductive and deductive reasoning can be used to teach it to adapt to different student behaviors. (5)
 - A)
 - B) **Evaluate** the performance of informed search over uninformed search when solving a constraint satisfaction problem like Sudoku. (5)
- 2) **Analyze** how Alpha-Beta pruning improves efficiency in minimax trees for strategic decision-making in games like Tic-Tac-Toe. (5)
 - A)
 - B) Given the two propositional logic expressions:
Expression 1:
 $(P \wedge (P \rightarrow Q))$
Expression 2:
 $(P \wedge Q)$
Determine whether Expression 1 is **logically equivalent** to Expression 2. Use **logical equivalence laws** to **simplify** Expression 1. (5)
- 3) A banking expert system uses First Order Logic to verify loan eligibility. **Explain** how systematic inferencing supports this. (5)
 - A)
 - B) **Analyze** the need for ontological engineering in medical diagnosis systems that must adapt to different categories and defaults. (5)
- 4) In the Wumpus World, use Bayes' Rule to infer the location of a pit. Show how full joint distributions support this. (5)
 - A)
 - B) **Analyze** how semantic representation in Bayesian Networks affects the interpretation of conditional dependencies. (5)
- 5) **Assess** the role of utility theory in AI decision-making under uncertainty. (5)
 - A)
 - B) Analyse a decision network that models student decisions for choosing an online (5)

learning platform.

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